



Beyond the post: an SLR of enterprise artificial intelligence in social media

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Abstract

This study explores the impact of artificial intelligence (AI) on brand communication within corporate social networks, analyzing its benefits, ethical and technical challenges, and proposing responsible implementation strategies enriched with new theoretical contributions. To achieve this, a systematic literature review (SLR) was conducted based on the SPAR-4-SLR methodology by Paul et al. (2021), using 57 studies from Scopus and Web of Science over the past six years. This approach was complemented with recommendations from Kitchenham and Charters (2007) to ensure rigor and thoroughness in the analysis. The study reveals that artificial intelligence transforms interactions within corporate social networks by enabling effective personalization, optimizing customer experience, and enhancing satisfaction. Benefits include precise segmentation, predictive analytics, and customer service optimization through chatbots. However, significant ethical challenges also emerge, such as data privacy, algorithmic bias, and a lack of transparency in AI models. The necessity for responsible practices and regulations that foster user trust and mitigate risks associated with the implementation of AI in digital communication strategies is emphasized.

Keywords SLR · Artificial intelligence · Corporate social networks · Customer-company engagement · Digital marketing · Business communication

1 Introduction

Artificial intelligence (AI) has become a foundational pillar of digital transformation, presenting significant opportunities but also substantial challenges for organizations (Dondapati et al. 2022). In the domains of marketing and communication, the integration of AI enables the optimization of advertising strategies, the enhancement of customer experience, and the collection of critical data on consumer

preferences and behaviors (Huang and Rust 2021; Ullal et al. 2021; Villareal and Flor 2023).

For the purposes of this study, AI is defined as a set of advanced technologies, including natural language processing (NLP) and machine learning (ML) (Saravani et al. 2023), which enable companies to analyze large volumes of data generated on social networks (Binder et al. 2022) and to optimize personalized content in real time (Zeng et al. 2022). These technologies facilitate not only the identification of consumer behavior patterns but also the anticipation of future needs and the adaptation of interactions in real time, with the objective of maximizing customer satisfaction and operational efficiency (Dwivedi et al. 2021; Piotrowski 2023; Wang et al. 2023).

As AI processes data generated from social platforms, it supports personalized advertising strategies that increase social media conversion rates. Recent examples from platforms like Twitter and TripAdvisor illustrate AI's role in opinion analysis and brand reputation management (Ducange et al. 2019; Nazir et al. 2023; Sripath et al. 2018). However, these benefits are accompanied by significant

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ethical and technical challenges, including data privacy management and algorithmic biases, which may undermine consumer trust and the sustainability of business practices (Huang and Rust 2021; T       2021).

The widespread adoption of AI has also raised ethical concerns, such as privacy, algorithmic biases, and societal impacts. The scientific community has highlighted additional risks, including excessive consumerism and job displacement, underscoring the importance of ethical and responsible adoption (Cardon et al. 2023; Lowrey 2023; Rai 2020; Wu and Monfort 2023).

Despite the notable growth in research on AI applied to digital marketing, studies that holistically integrate the impacts of AI on business management, customer experience, and ethical challenges remain scarce. Recent systematic studies have highlighted the need to consolidate existing knowledge to enable effective AI applications in business environments (Chintalapati and Pandey 2022; Dobrucali and       2021; Dwivedi et al. 2023; Giuggioli and Pellegrini 2023; Haleem et al. 2022). Although digital marketing has been widely studied, there remains a gap in understanding its application specifically in corporate social networks, an area that has received less academic attention (Peltier et al. 2024; Naz and Kashif 2024).

To address these gaps, this study employs a systematic literature review (SLR) following the rigorous SPAR-4-SLR protocol proposed by Paul et al. (2021), inspired by the analytical approaches of Kitchenham and Charters (2007) and Kitchenham et al. (2009). This methodology structures each research phase around specific questions aimed at exploring how AI is transforming marketing practices on corporate social networks, the benefits and risks associated with its adoption, and effective strategies to mitigate these risks.

The research questions guiding this systematic review are as follows:

- RQ1: How is AI transforming corporate social media interactions, and what impact does this have on the customer experience?
- RQ2: What are the primary benefits associated with AI adoption in the context of corporate social media?
- RQ3: What are the risks associated with AI implementation in digital communication strategies, and how can these risks be mitigated?

These research questions are designed to provide a comprehensive and up-to-date understanding of AI implementation in corporate social media environments, emphasizing the importance of a structured methodological approach to address critical issues associated with advanced technology.

2 Overview

In this section, the role of AI in corporate social networks and its impact on digital marketing are examined, analyzing both the opportunities and risks associated with its implementation. The benefits of AI in personalization strategies, optimizing customer experience, and the use of predictive tools are highlighted, along with the capacity of chatbots to enhance customer service. Additionally, ethical risks, such as data privacy and algorithmic bias, are addressed, emphasizing the need for a balanced approach that considers both the benefits and limitations of AI in corporate communication.

2.1 AI in corporate social networks

The digital era has brought a proliferation of social media platforms, making them crucial tools for corporate strategies (Pangkey et al. 2019). Today, social networks have solidified not only as communication tools but as indispensable elements of the global business ecosystem (Villareal and Flor 2023). In parallel, artificial intelligence has emerged as a technology with the potential to enhance the effectiveness and efficiency of platform management and strategy (Nazir et al. 2023; Nesterenko 2021).

In particular, the integration of AI into social media has transformed how businesses interact with consumers and understand their behaviors (Sharma and Shafiq 2022). Advanced AI tools, such as ML and NLP are now essential for analyzing and responding to user dynamics in real time (Olujimi and Ade-Ibijola 2022). Thus, AI-powered recommendation systems use advanced algorithms to filter and suggest relevant content to users, enhancing retention and satisfaction (Covington et al. 2016). Additionally, NLP techniques, such as recurrent neural networks, enable more sophisticated sentiment and opinion analysis of user-generated text, which is vital for marketing and customer service strategies (Almuqren and Cristea 2023).

In the field of customer service, AI-powered chatbots have improved support efficiency and quality by simultaneously managing numerous customer interactions and providing personalized responses (Zheng et al. 2023). Predictive analytics models are also employed to anticipate consumer trends and behaviors, allowing companies to take a proactive approach to their marketing strategies (Sari et al. 2020).

While the importance of AI in analyzing and managing online social interactions is emerging as a critical component of a company's digital marketing strategy, its integration into social media adds new value. According to Morris (2023), the integration of AI in social media improves organizations' reputation and engagement.

While AI holds the promise of improving how businesses interact and communicate through social media, with market growth expected at an annual rate of 27% from 2020 to 2025 (Datacenter Dynamics 2022), it is crucial to recognize and address the associated risks. In this regard, the challenge lies in managing risks such as privacy breaches, data security, algorithmic bias, and a potential lack of transparency in processes (Binder et al. 2022; Church et al. 2023; Dwivedi et al. 2021; Wang et al. 2023). Túnéz (2021) emphasizes that unethical or mismanaged implementation of AI in social networks could damage companies' reputation and trust. In this context, it is essential to consider that risk management depends not only on technical artifacts but also on appropriate human judgment, as purely technical approaches may be insufficient amid increasing uncertainty (Crawford and Jabbour 2024). Thus, the European and Commission (2019) highlights the importance of balancing innovation with caution, suggesting the implementation of regulations and strategies for the ethical and responsible use of AI in this field.

2.2 Importance of corporate social networks in digital marketing

Social media has gained a central role in corporate strategies, becoming essential pillars of digital marketing. According to a report by Statista (2024), global spending on digital advertising reached USD 549.51 billion in 2022, and is projected to grow to USD 870.85 billion by 2027. In this context, social networks are fundamental for acquiring and retaining customers, thus supporting the marketing strategies of companies across various sectors.

Today, digital marketing is a core component of business strategies in the modern era (Ijomah et al. 2024). A report from Sprout Social (2023) indicates that 90% of business leaders believe their company's success depends on insights gained from social media to guide their strategy, and 80% of these leaders anticipate an increase in their social media budgets over the next three years.

The introduction of generative artificial intelligence in 2022 has further transformed this landscape (Kshetri et al. 2023). Since its emergence, the use of AI in content creation and customer service has increased notably, with a projected growth rate of 318% by 2024 in key activities (Hootsuite 2024). While AI offers new opportunities to enhance efficiency and personalization in digital marketing (Pangkey et al. 2019) and corporate social networks, it also faces certain challenges, such as the acceptance of AI-generated content, which is still met with skepticism by the public (Chen et al. 2023), and the implementation of strategies on social media (Al-Youbi and Mahnabi 2024).

Social media strategies are fundamental pillars for achieving brands' primary objectives today: brand acquisition,

loyalty, conversion, awareness, positioning, and reputation (Kotler et al. 2023). To this end, as stated by Tuten (2023), the four main groups of social media strategies are Social Community, Social Publishing, Social Entertainment, and Social Commerce. The first of these focuses on building relationships through continuous interaction, and AI tools, such as chatbots, enable simultaneous consultations, optimizing staff costs and operational efficiency. Regarding the strategy of creating and distributing engaging content—social publishing—AI facilitates the rapid creation of personalized audiovisual resources. As for the social entertainment strategy, AI is used to track the preferences and interests of the brand's target audience, attracting users who engage with compelling content while avoiding advertising stimuli. Finally, social commerce strategies, using AI, enable sales on social networks without the need to access any e-commerce platform. By integrating these strategies with AI, companies can not only improve their digital visibility but also strengthen customer relationships, transforming interaction into optimal tangible outcomes.

2.3 Benefits of AI in corporate social media strategies

The growing importance of social media in digital marketing is driven by the use of AI, which enables the optimization of these business strategies through various applications. One of the most significant advances is personalization at scale, as this technology allows brands to create unique user experiences by analyzing data and anticipating individual needs, which enhances the accuracy of audience segmentation and the relevance of content (Tinkler 2023). This not only improves the interaction between the company and the customer but also increases conversion rates and customer loyalty.

A second benefit is the optimization of customer service management. AI-powered chatbots and virtual assistants can handle countless customer queries simultaneously, providing real-time responses and continuously learning from interactions to improve service (Zheng et al. 2023). This capability enhances operational efficiency and customer satisfaction, which are key elements in a highly competitive market environment.

In addition, AI facilitates predictive analytics in social media, allowing companies to anticipate consumer trends and behaviors. This is essential for content planning, product launches, and designing more effective marketing campaigns (Benabdelouahed and Dakouan 2020). For example, AI can identify behavioral patterns in real time, enabling companies to quickly adjust their strategies to maximize impact and return on investment.

Finally, automating sentiment analysis is another key benefit of AI. AI tools can analyze comments and social media posts to gauge audience perception and emotional responses to brands or products. This enables companies to respond more effectively to crises or manage issues before they escalate (Varsha et al. 2021). Overall, AI in social media empowers businesses with improved efficiency, richer interactions, and deeper data insights for strategic decision-making.

2.4 Challenges and risks of AI in corporate social networks

Despite its benefits, the implementation of AI in social networks also faces significant challenges, especially in terms of privacy, data security, algorithmic bias and lack of transparency. In terms of privacy and data security breaches, the massive collection of information can lead to misuse or exposure of sensitive data, affecting both users and organizations (Misra and Tiwari 2022). AI algorithms, which require access to large volumes of data, may inadvertently violate users' privacy or generate profiles without explicit consent (Wang et al. 2023). Furthermore, Al-Rubaie and Chang (2018) also point to the possibility of compromising private data, even when anonymized, and highlight the opacity in machine learning models, which poses significant challenges for transparency and trust in AI applications.

Another critical challenge is algorithmic bias. Algorithms used in content curation and personalization of the user experience can perpetuate existing biases, resulting in amplified discrimination or stereotypes on social media platforms (Deldjoo et al. 2023). This is because algorithms often reflect biases present in training data. An example of this is how search engines can reinforce negative stereotypes and discrimination (Noble 2018), a problem that is potentially magnified on social media platforms.

Finally, in relation to the 'lack of transparency' risk, the opacity of algorithms in AI systems is another traditional area of concern (Burrell 2016). The 'black box' of AI makes it difficult to understand how algorithmic decisions are made, affecting accountability, user trust (Sidlauskiene 2022) and difficulties in regulating these technologies (Cath 2018). Efforts to counter these challenges have led to the development of more ethical and transparent AI practices, seeking to establish mechanisms that allow for the auditing and explanation of algorithmic decisions (Franzoni 2023).

3 Methodology

To address the research questions, this paper employs a systematic approach to reviewing the existing literature, ensuring a comprehensive and methodologically rigorous analysis. Systematic Literature Reviews are particularly valuable in emerging and dynamic fields, such as artificial intelligence applied to corporate social networks, as they enable the integration and synthesis of knowledge in a structured way, identifying gaps and inconsistencies that might go unnoticed with other approaches (Paul and Criado 2020). To achieve this goal, we follow a rigorous protocol based on the 'Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR)' approach proposed by Paul et al. (2021), which guides all phases of the process, from the identification of relevant literature to the evaluation and synthesis of findings.

The SPAR-4-SLR protocol provides a framework based on four fundamental components: scientific procedures, selection rationales, quality assessment, and review synthesis. These components ensure that each phase of the review is organized and aligned with the study's objectives, allowing us to rigorously identify, evaluate, and synthesize relevant literature. Furthermore, we align with the recommendations established by Kitchenham et al. (2009) and Kitchenham and Charters (2007), ensuring scientific rigor at each stage of the review's development. These phases provide an additional structure, complementing SPAR-4-SLR in the identification of studies and the categorization of findings as shown in Fig. 1.

Recent studies have demonstrated that the systematic approach of SLR is particularly effective in detecting research gaps that may remain unnoticed in less structured reviews, such as critical or narrative reviews, thereby advancing knowledge in a more grounded and reliable manner (Kraus et al. 2021; Paul et al. 2021). Additionally, it provides a clear framework for the replicability and comparability of future studies, thus strengthening its value as an essential methodological tool in academic research (Durach et al. 2017).

3.1 Assembling

This study aims to thoroughly explore and analyze the role of AI in corporate social networking strategies by identifying and synthesizing its applications, benefits, challenges, and associated risks. The ultimate objective is to enhance corporate decision-making and marketing strategies, while also contributing new theoretical insights for future research.

Through the literature review, several key research gaps were identified, leading to the formulation of the following research questions, which are designed to provide a

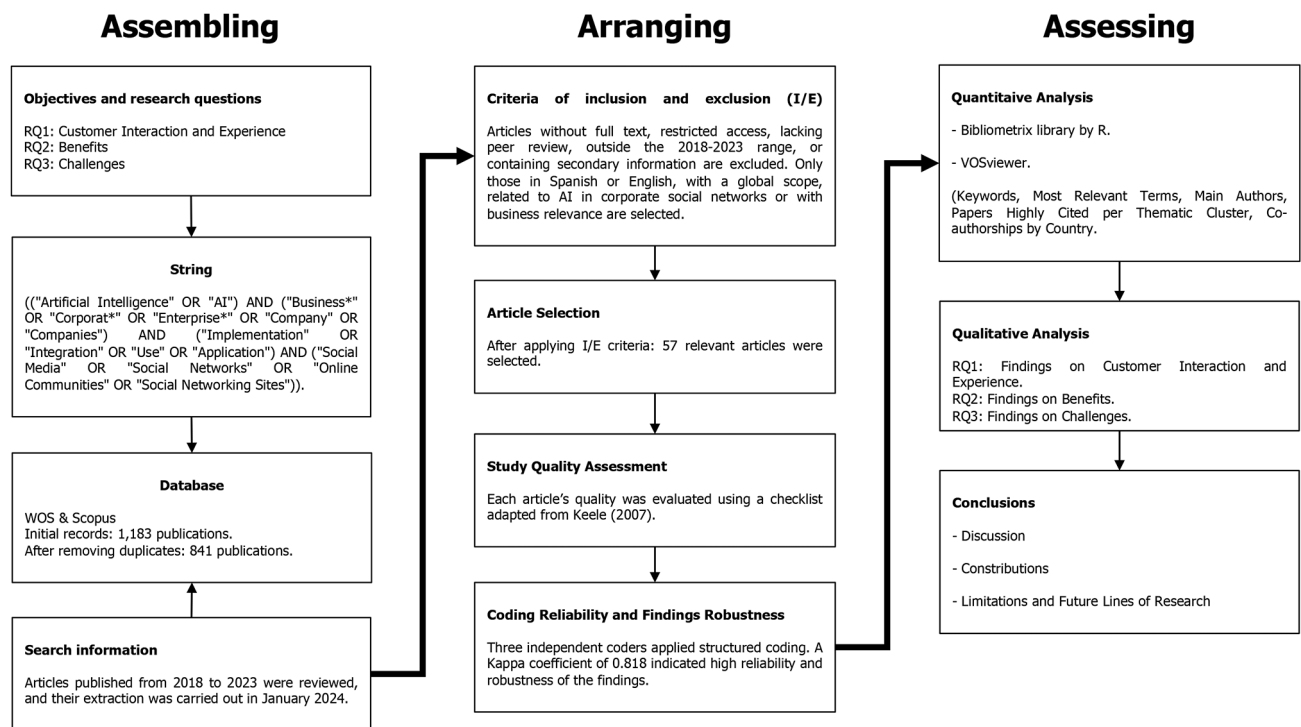


Fig. 1 Stages of SLR adapted from Paul et al. (2021)

Table 1 List of research questions for SLR

Nº	Research Questions	Associated Motivation
RQ1	How is AI transforming interaction on corporate social networks, and what impact does this have on customer experience?	Explore how AI has redefined communication between companies and audiences on social media, enhancing customer engagement and satisfaction, which are essential for corporate success (Chintalapati and Pandey 2022; Zheng et al. 2023).
RQ2	What are the main benefits associated with adopting AI in the context of corporate social networks?	Assess the overall advantages of AI in the corporate social media landscape, identifying key areas of competitive advantage and operational efficiency (Nazir et al. 2023; Sun et al. 2022).
RQ3	What risks are associated with implementing AI in digital communication strategies, and how can these risks be mitigated?	Understand the primary risks associated with AI usage, including ethical and privacy concerns, and examine the strategies currently applied or recommended to ensure safe and ethical corporate practices (Cardon et al. 2023; Rai 2020).

comprehensive understanding of AI's impact on corporate social networks as shown in Table 1.

Following the SPAR-4-SLR protocol outlined by Paul et al. (2021), a structured search strategy was developed to ensure the identification and acquisition of relevant studies. Web of Science (WOS) and Scopus were selected as the primary databases due to their extensive coverage of

high-quality, peer-reviewed literature in fields relevant to artificial intelligence and corporate social media.

To align with the research questions, a specific search string was created using keywords such as “artificial intelligence,” “businesses,” “implementation,” “social network,” and “social media.” Boolean operators “OR” and “AND” were employed to refine the search, with terms derived from prior research and thesauri to ensure both relevance and comprehensiveness (Kitchenham and Charters 2007; Kitchenham et al. 2009). This search was limited to studies published between 2018 and 2023 and included publications in Spanish and English to reflect the prevalence of these languages in the scientific literature. The complete string is shown in Fig. 2.

The initial search yielded 1,183 publications. After duplicates were removed using Mendeley, the number was reduced to 841 unique records. These records were then assessed for relevance based on the research scope.

This assembling phase established a structured foundation for the subsequent steps, facilitating a rigorous and methodical literature selection process as recommended by Paul et al. (2021). The identified studies would next undergo detailed screening, applying inclusion and exclusion criteria to ensure alignment with the study's objectives and quality standards.

String: (("Artificial Intelligence" OR "AI") AND ("Business*" OR "Corporat*" OR "Enterprise*" OR "Company" OR "Companies") AND ("Implementation" OR "Integration" OR "Use" OR "Application") AND ("Social Media" OR "Social Networks" OR "Online Communities" OR "Social Networking Sites"))

Fig. 2 SLR search string

Table 2 Criteria of inclusion and exclusion (I/E)

Criterion	Inclusion	Exclusion
Originality	Primary information, avoiding interpretations or summaries of previous works.	Inaccessible articles, limited content, or not meeting peer-review standards.
Publication Period	Works published between 2018 and 2023 to ensure data currency.	Articles published outside the range 2018 to November 2023.
Availability	Full-text, open-access articles to enable comprehensive information collection.	Articles not available in full text or with restricted access.
Geographic Scope	No geographical limits, aiming for a global perspective.	Articles that do not fit the global scope of the study.
Language	Articles in Spanish or English, reflecting the prevalence of these languages in scientific publication.	Publications in languages other than Spanish or English.
Thematic Relevance	Publications related to the implementation, integration, or use of AI in corporate social networks, and demonstrating relevance to the business sector.	Articles unrelated to the implementation, integration, or utilization of AI in corporate social networks, or lacking relevance to the business sector.
Literature Type	Scientific articles meeting rigor and peer-review standards.	Books, book chapters, experience reports, grey literature, and other non-article publications not meeting scientific standards.
Duplication	-	Duplicate publications to avoid redundancy and bias.

3.2 Arranging

At this phase, the objective was to organize and refine the collected literature by applying inclusion and exclusion criteria, conducting a detailed selection of articles, assessing the quality of the selected studies, and implementing a coding process to ensure consistency in data interpretation.

3.2.1 Criteria of inclusion and exclusion

To focus this systematic review on relevant studies, specific inclusion and exclusion criteria were defined, as shown in Table 2. These criteria were designed to ensure that the selected articles were current, accessible, and directly relevant to the study's objectives. The inclusion criteria encompassed original publications with primary information, avoiding interpretations or summaries of previous work. Additionally, only open-access articles in full text were included, in line with the commitment to promote open science (Paul et al. 2021; Fatmawati et al. 2023).

The review focused on articles published between 2018 and 2023 to ensure data currency. Publications in Spanish and English were included, with no geographical restrictions, allowing for a global perspective. Priority was given to studies focusing on the implementation, integration, or use of artificial intelligence in corporate social networks that demonstrated relevance to the business sector (Kitchenham 2004; Dybå and Dingsøyr 2008; Paul et al. 2021). Conversely, inaccessible articles, those with limited content, non-scientific publications, duplicates, and studies outside the defined time range were excluded.

Although the inclusion of grey literature could provide additional information on the application of artificial intelligence in corporate social networks, we chose to exclude it in this study. This decision was based on the need to maintain a high standard of academic rigor and reliability, prioritizing peer-reviewed sources that ensure the validity and quality of the findings (Adams et al. 2017; Mahood et al. 2014).

3.2.2 Article selection

The selection of articles followed a meticulous process based on the search string and the predefined inclusion/exclusion criteria. Initially, 1,183 publications were identified in the Scopus and WOS databases. After removing duplicates in the Mendeley reference manager, the number was reduced to 841 unique articles. Subsequently, the inclusion/exclusion criteria were applied through a review of titles, abstracts, and keywords, resulting in a preliminary selection of 284 studies.

Of these 284 studies, 176 did not meet the established criteria. The exclusion of these studies was carried out following a structured process, beginning with the removal of inaccessible articles or those whose full text was unavailable due to access restrictions in their respective journals, making adequate evaluation impossible (e.g., Abeki 2022; Agnihotri 2021). Secondly, studies were discarded if, although they addressed topics related to artificial intelligence, they did not focus on its implementation, integration, or use in corporate social networks nor demonstrated relevance to the business sector (e.g., Baresi and Tamburri 2023; Teplova et al. 2023). Finally, articles that provided an overview of AI techniques but lacked the specific focus or scientific rigor required for our systematic review were excluded (e.g., Jacobides et al. 2021; Lee and Oh 2019).

Articles considered for inclusion needed to address the implementation, integration, or use of artificial intelligence in corporate social networks and demonstrate their relevance to the business sector. Those that did not meet these criteria, as well as inaccessible articles, non-scientific publications (such as books or book chapters), articles irrelevant to the topic, duplicates, and those published outside the defined time range were excluded. Full access was obtained for the 71 preselected studies, and an in-depth reading of each text was conducted to confirm their relevance and quality. Articles that did not address the research questions posed were discarded, documenting the number and reason for each exclusion.

After this rigorous evaluation, 57 articles were selected for the final analysis. According to Paul et al. (2021), the total number of selected articles exceeded the average annual publications (9.5 articles per year), supporting the maturity of the topic domain and ensuring the feasibility of an adequate systematic review. Moreover, following the methodological recommendations of Booth et al. (2016), Petticrew and Roberts (2008), and Grant and Booth (2009), priority was given to selecting studies of high quality and relevance overachieving a fixed number of articles, given

the specific focus of this review. These authors emphasize that, in systematic reviews of specialized topics, a moderate number of studies can provide a comprehensive perspective as long as rigorous methodological criteria are applied. This approach ensures that the selected articles adequately represent the current state of research on the use of AI in corporate social networks.

3.2.3 Study quality assessment

To ensure the quality and thoroughness of the selected studies (Paul et al. 2021), each article was assessed based on a checklist adapted from Keele (2007), consisting of 10 questions, as shown in Table 3, related to the clarity of objectives, soundness of research design, appropriateness of data collection methods, focus of analysis, diversity of perspectives, among others.

Following the recommendation of Genc-Nayebi and Abran (2017), we selected studies that answered “yes” to at least 7 questions. This checklist ensures that the included studies contain relevant and valuable information for the purpose of our research. The entire quality assessment process guarantees the robustness of the studies on which this systematic review is based.

3.2.4 Coding and reliability assessment (Cohen’s Kappa)

To ensure accuracy in the qualitative analysis of the selected studies, a structured coding process was conducted involving three independent coders. This process began with the identification of predefined codes based on emerging categories derived from a preliminary review of the literature. The main codes included key themes such as “social media engagement,” “interaction personalization,” “market segmentation,” “data privacy,” and “customer satisfaction.” These labels were selected for their relevance to the implementation of artificial intelligence in corporate social networks and their impact on customer experience and business management.

With the idea of synthesizing, these codes were grouped into three broader categories: “Impact of AI on Company-Customer Interaction and Customer Experience,” “Overall Benefits of AI,” and “Challenges and Risk Management.” This structured approach allowed for consistent capture of the study’s key themes, ensuring that interpretations were coherent and aligned with the research objectives.

These codes were tested by 3 independent coders and to assess the reliability of the selections made by the coders, Cohen’s Kappa coefficient was applied, yielding a value of 0.818, which indicates an almost perfect agreement according to the Landis and Koch scale (1977). This high level of agreement ensures that the emergent categories in the

Table 3 Quality checklist Keele (2007)

Nº	Question
1	Are the aims of the study stated clearly?
2	Is the basis of evaluative appraisal clear?
3	How defensible is the research design?
4	Are data collection methods described adequately?
5	Has the approach to, and formulation of, analysis been conveyed adequately?
6	Has the diversity of perspectives and contexts been explored?
7	Are there any links between data, interpretation, and conclusions?
8	Is the reporting clear and coherent?
9	Has the research process been documented adequately?
10	Could the study be replicated?

qualitative analysis are consistent and reliable, minimizing subjective bias in data interpretation and reinforcing the robustness of the findings. In line with other methodological studies in systematic reviews, such as McHugh (2012), which highlights that a Kappa value above 0.80 is a strong indicator of reliability in qualitative coding, this result supports the robustness of our analysis outcomes.

3.3 Assessing

At this phase, the collected articles were analyzed, presenting the main findings. A mixed-methods approach was employed for the analysis. The first step involved conducting a quantitative analysis of the 57 selected articles, utilizing the Bibliometrix R package developed by Aria and Cuccurullo (2017), along with the VOSviewer tool to uncover patterns, trends, and key relationships within the existing literature. Through VOSviewer, a map of predominant keywords and their interconnections was extracted and visualized, along with a co-authorship network by country to illustrate collaborations and the geographic scope of the research. Additionally, a word cloud was generated with Bibliometrix, allowing observation of the prominence of various concepts within the studied corpus.

This quantitative approach was initially used to identify key themes and general patterns in the literature, providing a structured foundation that facilitated subsequent interpretation in the qualitative analysis. Thus, the quantitative analysis not only quantifies the impact and evolution of AI in corporate communication strategies—specifically in terms of the presence and relevance of topics and authors—but also guides the qualitative analysis toward areas of particular interest.

Subsequently, a qualitative content analysis was conducted, focusing on an in-depth examination of the topics covered to address the research questions. This combination of quantitative and qualitative analysis allows for the identification of emerging research lines and practices in the use of AI in corporate social networks, providing a comprehensive view of its development in the business domain.

The results obtained in this phase provide a solid foundation for the Discussion and Conclusion sections, where the findings will be interpreted within the context of the existing literature, exploring their theoretical and practical implications, and proposing future directions for research on AI applied to corporate social networks.

4 Results

This analysis combines quantitative and qualitative approaches to provide a comprehensive view of how AI is transforming social networking strategies in the corporate

environment. Initially, a quantitative analysis is presented to identify key patterns, trends, and connections within the existing literature, establishing a solid foundation on which the subsequent qualitative interpretation is based. Next, the qualitative analysis focuses on delving deeper into emerging themes, providing a more nuanced understanding of the benefits, challenges, and risks associated with implementing AI in social networks.

4.1 Quantitative analysis

The initial quantitative analysis provides a robust empirical basis for understanding the influence and current trends in the implementation of AI in corporate social networking strategies. By utilizing tools such as Bibliometrix and VOSviewer, key patterns, trends, and relationships within the reviewed literature are identified. This approach allows for mapping the impact and evolution of AI in corporate communication, as well as highlighting the main areas of international collaboration, predominant themes, and the most influential authors and studies.

4.1.1 Keywords

In the quantitative analysis of the presence and connections between keywords related to artificial intelligence in corporate social networks, as shown in Fig. 3, the systematic review highlights the predominance of the keywords “social media” (25 occurrences, 41 total link strength), followed closely by “artificial intelligence” (22 occurrences, 35 total link strength). These two terms form the central core of the research, reflecting their defining role in the existing literature.

On the other hand, the keywords “deep learning” and “social networking (online)” emerge as areas of specialized interest within the field, with “sentiment analysis” closely linked to artificial intelligence, underlining its relevance in assessing companies’ perceptions and managing customer experience. Similarly, the data reveals that machine learning and natural language processing are fundamental pillars in the understanding and analysis of big data, which is crucial for the development of digital marketing strategies. “Management” and “Facebook” appear as terms with lower occurrence but still relevant, suggesting a focus on the practical application and management of AI in corporate social contexts.

These findings indicate a clear trend towards the adoption of advanced AI technologies for data analysis in social networks and their application in digital marketing strategies and managerial decision-making.

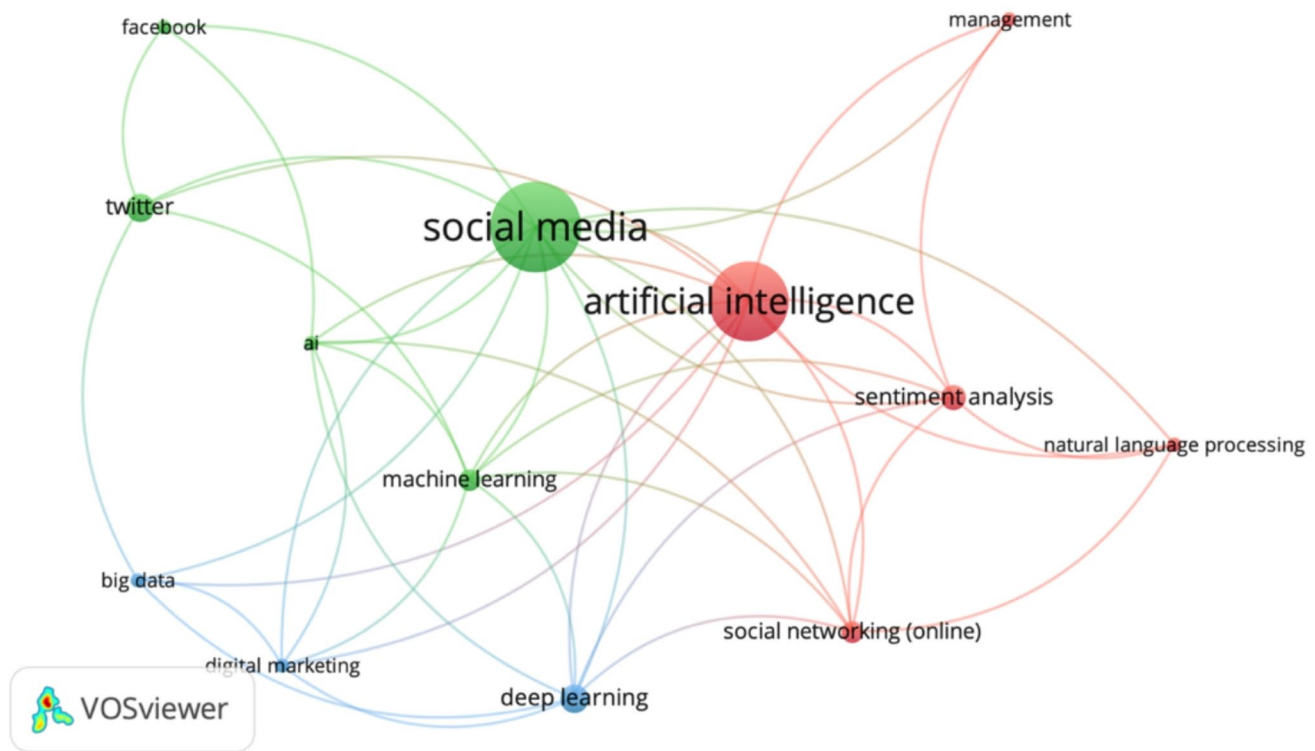


Fig. 3 Main Keywords

4.1.2 Most relevant terms

The dominant term associated with AI in corporate social networks is “social media” underscoring its centrality in the field (see Fig. 4). “Sentiment analysis” and ‘artificial intelligence’ also stand out, indicating their importance as areas of study and application within this context. Other terms such as “natural language processing”, “big data”, “deep learning” and “analytics” appear with considerable size, reflecting their relevance in the related literature. The presence of “Facebook” and “twitter” suggests that these specific platforms are crucial for research in this area.

On the other hand, the terms “behavior,” “impact,” “sales,” and “management” are also found, implying a focus on how AI in social networks affects consumer behavior, sales performance, and business management. These terms, along with “decision-making” and “knowledge,” illustrate the interconnection between AI technology and practical business functions, highlighting the depth and multidimensionality of research in this field.

4.1.3 Main authors

The most cited publication, as shown in Table 4, is the work of Dwivedi et al. (2021) being the most significant

contribution to the field of study. It is a much further step away, we find the other research where we can highlight those of Marsden et al. (2020), Obschonka et al. (2020) and Ducange et al. (2019).

The different disciplines of the journals where the most mentioned authors are published show the diversity of approaches and the importance of AI in different research contexts within corporate social networks, highlighting the multidisciplinary nature and influence of AI in improving the understanding and management of online interactions.

4.1.4 Papers highly cited per thematic cluster

The most cited papers focus on three major thematic clusters, Artificial Intelligence, Behavior and Emotions. Similarly, the most cited research in the corpus, “Setting the Future of Digital and Social Media Marketing Research” by Dwivedi et al. (2021) published in the “International Journal of Information Management”, has set a benchmark for the study of digital and social media marketing, reflecting its central importance with 427 citations (see Table 5). Other notable works include the analysis of the impact of AI-powered technologies on the purchasing decisions of Instagram users in the age of digitization and the study of the spread of misinformation by social bots in the media, each



Fig. 4 Cloud of words

Table 4 Main authors

Paper	Total Citations
Dwivedi YK, 2021	Int J Inform Manage 427
Marsden C, 2020	Comput Law Secur Rev 42
Obschonka M, 2020	Small Bus Econ 37
Ducange P, 2019	Eng Appl Artif Intel 36
Giannakis M, 2022	Ann Oper Res 27
Yeo SF, 2022	Technol Forecast Soc Change 24
Ameen N, 2022	Psychol Market 21
Johnson M, 2022	Int J Inform Manage 19
Vizoso A, 2021	Media Commun-Lisbon 19
Hajli N, 2022	Brit J Manage 16

highlighting key challenges and trends at the intersection of AI, society and consumer behavior. These papers emphasize the relevance of AI in generating insights from large volumes of data and in understanding the current dynamics of marketing and information management.

4.1.5 Co-authorships by country

International collaboration in artificial intelligence research applied to corporate social networks shows a diverse and

globalized picture. As shown in Fig. 5, England and the United States lead in terms of published papers and citations, indicating a strong impact on the scientific community. The significant representation of India highlights its growing contribution to the field. France and Australia also emerge as key contributors, with substantial numbers of citations, reflecting the influence of their research.

The co-authorship network reveals that England serves as a central node of collaboration, connecting with several European countries and beyond. The interconnection between countries such as Canada, India and the United States illustrates a strong and extensive collaborative network, supporting the advancement of AI knowledge in the context of corporate social networks.

4.2 Qualitative analysis

Having conducted the quantitative analysis and disentangled the key patterns and trends in the literature on AI in corporate social networks, the next step is to present the qualitative findings. This analysis focuses on answering the research questions identified in the study. A detailed

Table 5 Papers highly cited per Thematic

Title	Source	Year	Thematic Cluster	Total citation
Setting the future of digital and social media marketing research: perspectives and research propositions.	International Journal of Information Management	2021	Artificial Intelligence	427
Investigating the impact of ai-powered technologies on instagrammers' purchase decisions in digitalization era—a study of the fashion and apparel industry	Technological Forecasting and Social Change	2022	Artificial Intelligence	24
Social bots and the spread of disinformation in social media: the challenges of artificial intelligence	British Journal of Management	2022	Artificial Intelligence	16
Vulnerability in a tracked society: combining tracking and survey data to understand who gets targeted with what content	New Media & Society	2020	Behavior	15
Detecting pain points from user-generated social media posts using machine learning	Journal Of Interactive Marketing	2022	Artificial Intelligence	8
Exploring the sentiment of entrepreneurs on twitter	Plos One	2021	Emotions	3
Comparing content marketing strategies of digital brands using machine learning	Humanities & Social Sciences Communications	2023	Behavior	2

table (see Annex Table A1) summarizing the main findings derived from the 57 articles reviewed in this SLR is presented in the Annex, providing a comprehensive overview of the emerging categories and their relevance to the field of study.

In this **Results** section, Table 6 summarizes the emerging categories and the most significant qualitative findings, linking them directly to the research questions posed.

4.2.1 Transformation of interaction and its impact on customer experience (RQ1)

RQ1 explores how artificial intelligence (AI) is transforming the interaction between clients and companies on social media, enhancing personalization, reducing response times, and increasing customer satisfaction. These findings align directly with the study's goal of understanding how AI can optimize the user experience and strengthen company-client relationships.

Personalization at scale is one of the key benefits of AI on social media. Liu et al. (2023) highlight that AI enables advanced segmentation that maximizes content relevance, thereby increasing trust and effectiveness in social media marketing campaigns. This approach is particularly valued in industries like beauty, where Ameen et al. (2022) observe that AI and augmented reality (AR) enable personalized experiences that boost engagement and perceived value for users, improving satisfaction and brand loyalty.

In the customer service domain, the implementation of chatbots has proven to be an effective resource for replicating human interactions, providing immediate and personalized responses that strengthen the company-client relationship. Sari et al. (2020) and Kumar et al. (2023) note that chatbots not only enable continuous support but also significantly reduce wait times, enhancing the company's perceived proximity and improving customer satisfaction. Jansom et al. (2022) reinforces this point by demonstrating that interaction with these virtual assistants not only improves the customer experience but also increases the credibility of social media communication.

Real-time sentiment analysis is another powerful tool that AI brings to client-company interactions, facilitating the adaptation of communication based on emotions detected in customers. Salminen et al. (2022) and Labat et al. (2023) emphasize that emotional monitoring on social media enables companies to quickly identify potential crises or dissatisfaction, enabling proactive responses that enhance customer perception and prevent negative escalations. Similarly, Varsha et al. (2021) points out that sentiment analysis helps adjust messaging to align with the emotional state of the customer, creating more empathetic interactions that meet their expectations.

Moreover, AI optimizes content based on users' psychological needs, thereby increasing reach and visibility on social media. Zeng et al. (2022) observes that by tailoring content to respond to specific psychological motivations, AI fosters a deeper emotional connection, increasing engagement and customer loyalty. This emotionally oriented personalization capability, also supported by Tzafilkou et al. (2023) and Khan et al. (2020), contributes to an enriched

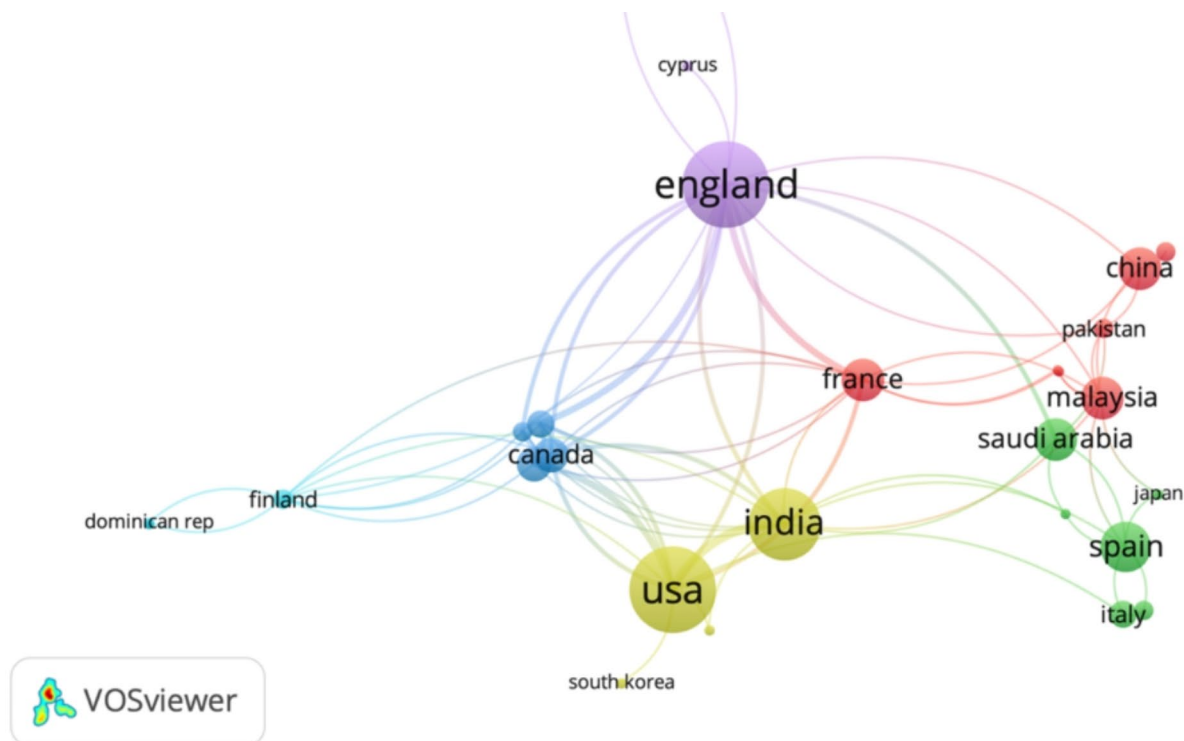


Fig. 5 Co-authorships by country

user experience, strengthening the long-term relationship between customer and company.

Finally, automating social media interactions through AI enables the management of large volumes of comments and inquiries in real-time, improving customer service efficiency. AI tools described by Olujimi and Ade-Ibijola (2022) and Rodrigo-Martín et al. (2021) allow for automatic and relevant responses, reducing wait times and reinforcing the perception of closeness to the brand. This not only increases customer satisfaction but also boosts loyalty and engagement, as customers feel valued and promptly attended to with accuracy and speed.

4.2.2 Benefits of adopting AI in corporate social networks (RQ2)

The implementation of AI in corporate social networks offers multiple benefits, notably in personalization, operational efficiency, and support for strategic decision-making. In terms of content optimization, AI enables SMEs to significantly improve their visibility and engagement by addressing users' psychological needs (Zeng et al. 2022). Additionally, AI-based technology enhances personalization on platforms like Instagram, increasing purchase intent and sales efficiency (Yeo et al. 2022).

Advanced segmentation is another key benefit. AI allows content to be adapted to specific audiences, maximizing its

relevance. Wang et al. (2023) and Ullal et al. (2021) note that AI facilitates precise segmentation based on users' past behavior, thereby improving user satisfaction and campaign impact. Similarly, Deldjoo et al. (2023) and Son et al. (2023) emphasize that AI-based recommendation systems enhance customer retention and enable more personalized CRM strategies, which increases customer loyalty.

AI has also strengthened emotion analysis and purchase intent detection. Studies such as those by Tzafilikou et al. (2023) and Sharma and Shafiq (2022) highlight the accuracy of AI in predicting purchase intent by analyzing emotions in food product campaigns and product forums, allowing timely adjustments to the offering. This ability to process large volumes of unstructured data on social media facilitates trend identification and real-time customer satisfaction monitoring (Chen 2023; Labat et al. 2023).

In customer service, AI enables continuous and personalized support through chatbots, which reduce operational costs and improve efficiency. Sari et al. (2020) and Kumar et al. (2023) underline that these systems provide quick and personalized responses, optimizing user experience and reinforcing the positive perception of the brand. Additionally, AI allows for the classification of specific customer problems and experiences, as noted by Salminen et al. (2022), generating valuable insights for continuous improvement and decision-making.

Table 6 Table 6

Nº	RESULTING CATEGORIES	DESCRIPTION OF THE FINDINGS	RQ ASSO-CIATED
1	Impact of AI on Business-Customer Interaction and Customer Experience	AI transforms interactions on corporate social networks by facilitating personalization and improving real-time response, enriching the customer experience through faster and more effective communication.	RQ1
2	Overall Benefits of AI	The primary benefits of AI adoption in corporate social media include improved content personalization, predictive analytics that optimize marketing campaigns, and the ability to segment audiences more effectively, increasing efficiency and ROI.	RQ2
3	Challenges and Risk Management.	Risks associated with the implementation of AI in digital communication strategies include ethical and privacy concerns, such as algorithmic bias and data manipulation. To mitigate these risks, it is suggested to establish clear ethical frameworks and regulations that protect users' rights.	RQ3

Real-time optimization of advertising campaigns is another key benefit of AI on social media. Ullal et al. (2021) and González-Briones et al. (2022) highlight that AI enables adjustments to ads based on user behavior, increasing return on investment and improving resource efficiency. Furthermore, Giannakis et al. (2022) and Dwivedi et al. (2021) emphasize that AI facilitates product development aligned with consumer demands through social media data analysis, accelerating the innovation process and product adaptation.

Finally, AI supports strategic decision-making by combining data analysis with human knowledge to identify complex patterns and extract valuable insights. Johnson et al. (2022) and Ducange et al. (2019) argue that AI enhances decision-making accuracy by enabling real-time data visualization, facilitating planning aligned with the company's strategic goals. In critical times, such as during pandemics, AI-based decision support systems are particularly useful in identifying real-time business opportunities, especially benefiting SMEs (Kumar et al. 2021).

4.2.3 Challenges and mitigation strategies in AI implementation (RQ3)

The implementation of AI in corporate social networks faces various ethical and technical challenges that impact the accuracy, transparency, and reliability of these systems. One of the most recurring issues is algorithmic bias, which can perpetuate stereotypes and affect fairness in decision-making. Marsden et al. (2020) and Raza et al. (2024) emphasize the importance of transparency and algorithm review to mitigate this bias. Szopa (2021) and Bol et al. (2020) highlight that algorithms can limit the authenticity of individual representations, creating predictable and homogeneous digital profiles, which raises ethical concerns. Deldjoo et al. (2023) also point out that recommendation systems require interdisciplinary approaches to address issues of fairness and transparency.

Privacy and security in the management of personal data represent another critical challenge. Studies by Liu et al. (2023) and Wang et al. (2023) indicate that intensive data collection can generate cognitive dissonance among users, especially younger ones, who perceive it as an invasion of their privacy. Almuqren and Cristea (2023) stress the need to improve the accuracy of AI models to avoid misunderstandings in interpreting user satisfaction, while Arora et al. (2023) recommend implementing authentication strategies and behavioral analysis to reduce cybersecurity risks.

The opacity of AI models, especially in the interpretation of automated decisions, also raises ethical and trust-related issues. Bol et al. (2020) and Dařena et al. (2018) explain that the lack of transparency in these models complicates the interpretation of results and presents ethical challenges regarding accountability. Johnson et al. (2022) and hlhoff & Willem (2023) suggest using explainable AI systems to facilitate understanding and promote transparency in corporate social media applications.

The complexity of managing large volumes of unstructured data is another notable challenge. Gao et al. (2023) and Vizoso et al. (2021) mention that variability in the quality and type of data on social media affects the accuracy of analyses, especially when attempting to capture emotions and opinions in real-time. This challenge is intensified on platforms like Twitter, where information is fast-paced and ambiguous, complicating immediate interpretation and response.

Finally, the detection of misinformation and bot control are essential for maintaining security and credibility on social networks. Hajli et al. (2022) and Marsden et al. (2020) warn that bots can manipulate public opinion and compromise the platform's integrity. Vizoso et al. (2021) and Church et al. (2023) highlight advancements in technologies like deepfakes and opinion polarization, suggesting

collaboration between platforms and the development of advanced tools to mitigate these risks.

5 Discussion

This SLR demonstrates how the integration of AI into corporate social networks is profoundly transforming business strategies, redefining communication between companies and audiences, and reshaping digital marketing strategies. The findings underscore the need for a balanced approach that not only leverages AI's transformative opportunities but also considers human oversight, ethical considerations, and operational demands to ensure a responsible impact on the digital economy.

5.1 Findings from the quantitative analysis

The quantitative analysis in this SLR reveals a series of key themes and frequent associations between terms like “artificial intelligence,” “social media,” “natural language processing (NLP),” “machine learning (ML),” and “big data.” These conceptual nodes highlight the centrality of automation in corporate social networks, driven by these advanced technologies. However, intensive automation raises significant concerns regarding algorithmic bias, previously noted in the literature by authors such as Burrell (2016), Noble (2018), and Rai (2020), who warn about the impact of these biases on perceptions of fairness in interactions, affecting user trust.

The correlation between terms such as “transparency,” “human oversight,” and “risk mitigation” in the quantitative analysis suggests a consensus in the scientific community on the importance of AI management that includes human oversight to promote transparency and accountability. This consensus supports the arguments of authors like Binder et al. (2022) and Ducange et al. (2019), who emphasize that despite technological advancements, human intervention remains essential for the responsible and ethical management of large volumes of data in corporate social networks.

5.2 Customer-company interaction and user experience (RQ1)

To answer RQ1, which examines how AI is transforming interactions on corporate social networks and its impact on customer experience, the findings indicate that personalization, sentiment analysis, and real-time interaction are key elements. These insights, based on the literature and studies reviewed, demonstrate that AI facilitates greater customer satisfaction by accurately adapting to users' needs and expectations on social media (see Sect. 4.2.1).

5.2.1 Personalization and enhancement of user experience

Personalization is one of the most significant advancements in user experience on social media. Tinkler (2023) points out that AI enables brands to create unique experiences by analyzing data and anticipating individual needs, which increases content relevance and improves conversion rates and customer loyalty. This finding aligns with Ameen et al. (2022), who report that AI and augmented reality (AR) in beauty brand applications foster personalized experiences that heighten user satisfaction, particularly among Generation Z, a group that highly values personalization and content relevance.

Dwivedi et al. (2021) also highlight that AI enables real-time optimization of campaigns and customization of the customer experience, though they caution that poor implementation can negatively affect experience quality. AI's ability to detect specific user needs and emotions, as noted by Salminen et al. (2022) and Khan et al. (2020), allows companies to tailor their messages and products to individual preferences, enhancing customer perception and strengthening the client-company relationship.

5.2.2 Optimization of real-time interaction and customer satisfaction

The literature and SLR findings also underscore the role of chatbots and virtual assistants in improving real-time interaction. Zheng et al. (2023) and Sari et al. (2020) explain how chatbots can handle multiple interactions simultaneously, providing instant and personalized responses, which enhances operational efficiency and customer satisfaction. This benefit is also observed in the study by Jansom et al. (2022), which notes that chatbots improve customer satisfaction through personalized interactions, efficiently resolving issues while keeping up with current trends.

Johnson et al. (2022) introduce the concept of “Informed AI,” which integrates human knowledge to improve the accuracy and interpretation of AI on social networks, enabling quick identification of quality issues in support conversations. This approach reduces wait times and increases customer satisfaction by facilitating interactions that are more precise and tailored to their needs. Kumar et al. (2023) reinforce this idea by highlighting how AI in social media enables customer service that emulates human interactions, thereby enhancing perceptions of closeness and trust.

5.2.3 Sentiment analysis and better understanding of customer needs

Sentiment analysis, powered by AI, also plays a fundamental role in enhancing client-company interactions. Varsha et

al. (2021) notes that AI tools allow companies to analyze comments and posts on social media to assess customer perceptions and emotions, facilitating a proactive and appropriate response to potential crises. Labat et al. (2023) and Bhargav and Arora (2022) demonstrate how AI enables real-time emotion detection and provides insights into public opinion, helping companies adjust their communication strategies to improve customer satisfaction and experience.

Salminen et al. (2022) and Tran et al. (2021) also emphasize how AI helps companies better understand emotions and specific aspects of products or services that are important to users. This ability to extract deep insights from unstructured comments allows companies to tailor interactions and campaigns to better meet customer expectations, thereby strengthening relationships and improving user experience.

5.2.4 Enhancing user satisfaction and trust

Finally, the responsible use of AI is essential for increasing user trust in brand interactions. Liu et al. (2023) highlight that responsible AI can enhance the effectiveness of social media marketing initiatives, boosting user trust and satisfaction. This is consistent with the findings of Ducange et al. (2019) and Hajli et al. (2022), who note that social bots can influence customer perception by increasing the visibility of both positive and negative information on social media. AI's capacity to manage large volumes of data and extract useful insights in real time allows companies to adjust their messaging and strengthen customer trust in the brand.

5.3 Benefits of adopting AI in corporate social networks (RQ2)

Regarding RQ2, the benefits of AI in corporate social networks include content optimization, advanced segmentation, predictive analysis, and support for strategic decision-making. These results underscore AI's positive impact on the effectiveness of digital strategies, maximizing content relevance and improving user satisfaction (see Sect. 4.2.2).

5.3.1 Personalization and precise segmentation

Personalization is one of the most prominent benefits of AI in social networks. Tinkler (2023) describes large-scale personalization as essential for enhancing customer interaction and loyalty by enabling brands to analyze data and anticipate individual needs. This finding is supported by Liu et al. (2023), who observe that AI improves personalization and efficiency in information management, optimizing communication and user satisfaction. Similarly, Zeng et al. (2022) highlights that AI-driven content optimization enhances social media visibility, a finding consistent with

the personalization and precise segmentation that Ullal et al. (2021) also describe in terms of tailored recommendations based on past user behavior.

Ameen et al. (2022) examine the impact of AI and augmented reality (AR) on personalized experiences, especially in the beauty industry, facilitating an emotional connection with the brand. This type of personalization enhances customer loyalty by allowing companies to meet specific audience needs and preferences, a benefit that aligns with Olujimi & Ade-Ibijola's (2022) analysis on how AI helps manage large volumes of interactions in an automated and personalized manner.

5.3.2 Optimization of customer service and improvement in satisfaction

The use of chatbots and virtual assistants significantly enhances customer service capabilities, a feature highlighted in studies by Zheng et al. (2023) and Sari et al. (2020). AI enables the simultaneous handling of multiple interactions, providing real-time responses that boost operational efficiency and customer satisfaction. Jansom et al. (2022) and Kumar et al. (2023) observe that AI-powered chatbots not only improve service efficiency but also help resolve common issues, increasing users' sense of closeness and optimizing their overall experience.

In this regard, Salminen et al. (2022) describe how AI models allow companies to analyze large volumes of user-generated content, facilitating the identification of specific issues and enhancing the customer experience. This observation aligns with Carvalho et al. (2019), who illustrate AI's use in extracting consumer sentiments and emotions in corporate social responsibility campaigns, demonstrating how AI can optimize customer service by enabling companies to better understand their customers' emotions and concerns.

5.3.3 Predictive analysis to anticipate trends

Another key benefit of AI in social networks is its ability to conduct predictive analysis, allowing companies to anticipate demands and adjust their strategies proactively. Benaïdelouahed and Dakouan (2020) emphasize that AI helps identify behavioral patterns in real time, enabling quick responses that maximize the impact of marketing strategies. Gao et al. (2023) support this advantage by developing a predictive model that facilitates real-time data-driven decisions, optimizing product and service planning.

Dwivedi et al. (2021) and Giannakis et al. (2022) also mention that AI allows for campaign optimization and product development through the analysis of sentiments and opinions on social media. The ability to analyze emotions and needs enables companies to adjust their strategies more

precisely, enhancing customer interactions and improving the overall experience.

5.3.4 Enhanced decision-making and operational efficiency

AI also facilitates decision-making through real-time analysis of trends and opinions. Ducange et al. (2019) and Chen (2023) highlight that AI-driven decision support systems allow for real-time reporting and the identification of positive or negative trends. This enables marketing managers to make informed decisions and adjust their campaigns based on recent data, thus optimizing operational efficiency.

Giannakis et al. (2022) explore how AI, through text analysis on social media, supports the new product development (NPD) process by enabling quick, cost-effective decisions based on shifting demands. This aligns with findings by Obschonka et al. (2020), who indicate that AI and big data can even estimate regional personality traits and entrepreneurial culture, which is highly useful for adapting strategies to specific contexts and enhancing the effectiveness of business decisions.

5.4 Challenges and mitigation strategies in AI implementation (RQ3)

Regarding RQ3, which investigates the risks of implementing AI in digital communication strategies, the findings highlight several critical ethical and technical challenges, including data privacy, algorithmic bias, and lack of transparency in models. These findings emphasize the importance of implementing responsible practices and promoting transparency to mitigate these risks and preserve user trust (see Sect. 4.2.3).

These results are consistent with prior literature that warns of the dangers associated with massive data collection and the opacity of algorithms, which can undermine user trust and negatively impact corporate reputation (Huang and Rust 2022; T    ez 2021).

5.4.1 Data privacy and security

Several studies in the table (e.g., Hajli et al. (2022) and Alshamari (2023) underscore the importance of protecting data privacy, aligning with risks noted by Al-Rubaie and Chang (2018) in previous literature, who warn that extensive data collection for algorithmic training may lead to privacy breaches, even with anonymized data. Misra and Tiwari (2022) reinforce this idea, suggesting that organizations should implement rigorous controls to prevent misuse of sensitive data.

5.4.2 Algorithmic bias and discrimination

Algorithmic bias is a recurring challenge in AI analysis on social media. Studies like Johnson et al. (2022) and Dwivedi et al. (2021) in the table highlight the risks of biases in AI models and propose mitigation strategies, such as integrating human knowledge and explicit labeling to reduce biases. This is consistent with observations by Deldjoo et al. (2023) and Noble (2018), who emphasize that algorithms can amplify stereotypes and perpetuate discrimination due to inherent biases in training data. To mitigate these issues, the literature suggests practices like algorithmic auditing and greater transparency in audience selection processes (Bol et al. 2020; Raza et al. 2024).

5.4.3 Model opacity and lack of transparency

The lack of transparency or “black box” nature of AI models was identified as a critical issue both in the table (e.g., Stockinger et al. (2023) and hlhoff & Willem (2023) and in prior literature (Burrell 2016; Sidlauskiene 2022). Studies propose solutions such as adopting explainable AI (XAI) models and labeling algorithmic decisions to facilitate the understanding and regulation of these systems (Franzoni 2023). This is directly related to mitigation strategies proposed in the literature, which include implementing linguistic rules to audit algorithms and ensure compliance with regulations like the GDPR (Binder et al. 2022).

5.4.4 Strategies for mitigating ethical challenges

The studies in the table suggest various mitigation strategies to address these challenges. For instance, Arora et al. (2023) recommend user behavior analysis and secure development protocols to mitigate cybersecurity risks in chatbots. These recommendations align with calls for increased regulation and control over AI usage in social media, as mentioned in the literature (European and Commission 2019; Church et al. 2023). Furthermore, Tran et al. (2021) and Khan et al. (2020) propose using advanced preprocessing and adaptable classification techniques to reduce interpretation errors in AI models, a strategy aligned with recommendations by Wang et al. (2023) to improve data processing accuracy in social environments.

5.5 Theoretical contributions and practical implications

This work represents the first systematic literature review (SLR) to comprehensively examine the role of artificial intelligence (AI) in brand communication within corporate social networks. By focusing on corporate strategies for

engaging with customers in these digital environments, this review not only synthesizes the practical benefits of AI—such as content personalization and predictive analytics—but also identifies new areas of opportunity and research gaps for further exploration by the scientific community. This study provides significant theoretical value by consolidating dispersed perspectives into a coherent framework, highlighting how AI enhances customer interaction, supports strategic decision-making, and introduces unique ethical and operational challenges. The implications of these findings extend beyond academic contributions, offering practical guidelines for practitioners seeking to effectively leverage AI within corporate social networks. The research questions (RQs) guiding this review provide a structured approach to understanding AI's transformative role, its potential benefits, and associated risks, shaping both future research and practical applications.

5.5.1 Theoretical contributions and practical implications by research question

RQ1: The Impact of AI on Corporate Social Media Interactions and Customer Experience.

The findings for RQ1 highlight the transformative impact of AI in enhancing customer interactions through personalization, sentiment analysis, and real-time engagement. From a theoretical perspective, this review contributes to understanding large-scale personalization as an essential mechanism to strengthen the brand-customer connection, underscoring AI's ability to respond to users' emotional needs on social media (Tinkler 2023; Salminen et al. 2022; Zeng et al. 2022). In practical terms, these findings suggest that companies adopting AI should prioritize tools that foster emotional engagement, as real-time sentiment analysis and interactions via chatbots have been shown to increase customer satisfaction and loyalty (Sari et al. 2020; Kumar et al. 2023; Jansom et al. 2022). For practitioners, integrating AI solutions that enable data-driven, responsive adjustments in communication strategies can significantly improve customer perception and reinforce brand loyalty, highlighting the importance of receptivity and empathy in enhancing the digital customer experience.

RQ2: Benefits of AI Adoption in Corporate Social Media Strategies.

The analysis of RQ2 identifies several benefits of AI in corporate social media strategies, including content personalization, predictive analysis, and advanced segmentation. Theoretically, this research provides a comprehensive view of AI's role in enhancing operational efficiency and strategic alignment, contributing to the literature by detailing how AI-based tools optimize resource allocation and maximize content relevance (Dwivedi et al. 2021; Wang et al. 2023;

Ameen et al. 2022). Practically, these conclusions suggest that companies can implement AI solutions enabling continuous learning from customer data, thereby facilitating more personalized and effective marketing strategies (Zeng et al. 2022; Tinkler 2023; Ullal et al. 2021). By using AI to anticipate user preferences and adjust content in real-time, companies can significantly improve engagement and conversion rates. For practitioners, these findings emphasize the importance of advanced data analytics and machine learning tools in optimizing customer understanding, which is crucial for gaining a competitive edge in the fast-paced digital environment.

RQ3: Risks and Mitigation Strategies Associated with AI in Digital Communication.

Findings related to RQ3 reveal the ethical and technical challenges associated with AI adoption in corporate social media, such as data privacy, algorithmic bias, and model opacity. Theoretically, this study contributes to the emerging discourse on AI ethics in digital communication, highlighting risks that may compromise consumer trust and corporate reputation (Binder et al. 2022; Túnuez 2021; Liu et al. 2023). Documenting these risks extends existing frameworks on the ethical use of AI, urging for more responsible and transparent practices in social media contexts (Marsden et al. 2020; Noble 2018). Practically, the findings suggest that companies should adopt robust governance frameworks prioritizing data privacy and bias mitigation, with transparency measures such as explainable AI to enhance user trust and support regulatory compliance (Franzoni 2023; Mühlhoff and Willem 2023; Arora et al. 2023). For practitioners, these results underscore the importance of continuous evaluation of AI models to prevent bias and uphold ethical standards, reminding that technical solutions must be accompanied by ethical oversight to protect brand reputation and maintain user trust in digital interactions.

5.6 Limitations and future lines of research

This study identifies both the advantages and critical areas requiring further investigation related to corporate communication using AI in social networks. While representing the first SLR in this area, the study has certain limitations that open up new opportunities for future research.

Firstly, this review focuses on open-access articles in English and Spanish, sourced from WOS and Scopus databases. Although this selection ensures a high-quality standard, it may restrict the diversity of perspectives. Expanding the analysis to other databases and additional publication types could provide a broader view of AI's impact on social networks, enriching understanding of the global context. Additionally, the rapid advancement of AI technologies presents a challenge in capturing the most recent

developments in this field (Nesterenko 2021). It is advisable for future research to periodically update this review to reflect the sector's ongoing evolution and capture emerging technologies with practical and ethical implications for digital communication.

5.6.1 Limitations and future research directions by research question

RQ1: Further Examination of AI's Impact on Social Media Interactions and Customer Experience.

For RQ1, future research could address the following key questions:

- How does real-time AI-based personalization impact customer perceptions of brand authenticity and trust on social networks? While chatbots and virtual assistants have been found to improve customer satisfaction (Sari et al. 2020), detailed analyses are needed to understand how these mechanisms might affect perceptions of closeness compared to human interactions.
- What are the psychological and emotional effects on consumers interacting with brands that use AI to analyze and respond to their emotions on social networks? Studies focused on sentiment analysis and real-time message adaptation (Zeng et al. 2022; Salminen et al. 2022) could provide a framework for deeper insights into how these technologies influence customer experience.

RQ2: Exploring New AI Applications in Social Media Strategy Optimization.

For RQ2, there are several areas where research could advance the applications of AI in social media strategies:

- How does the effectiveness of advanced segmentation and predictive analytics vary by industry and organization size? Although personalized segmentation has proven beneficial in some sectors (Wang et al. 2023), comparative studies are needed to explore which sector-specific or cultural factors influence AI adoption and effectiveness in social media.
- To what extent do advanced personalization and behavior prediction through AI affect purchase intent and loyalty compared to traditional methods? Studies that measure these variables in real campaigns could help identify the specific AI elements that significantly impact consumer behavior.

RQ3: Mitigating Ethical and Technical Risks in AI Adoption for Digital Communication.

For RQ3, which focuses on ethical and technical challenges, future research could delve into:

- What audit and AI evaluation methods are effective in mitigating algorithmic bias and increasing transparency on social networks? Although the need for explainable AI is recognized (Franzoni 2023; Mühlhoff and Willem 2023), empirical studies comparing different audit techniques in the social media context are lacking.
- How do regulatory frameworks and privacy policies affect AI development and implementation on social networks? Longitudinal studies analyzing the impact of specific regulations on AI applied to brand communication could provide valuable insights to align these tools with data protection and ethical standards.

Despite its limitations, this study lays an important foundation for understanding the use of AI in brand communication on social media. The proposed future research directions offer a clear path for enriching both theoretical knowledge and practical recommendations, fostering AI adoption that aligns with ethical principles and protects user privacy.

6 Conclusion

This study provides an in-depth understanding of how artificial intelligence (AI) is redefining communication within corporate social networks. Key benefits identified include personalized customer experiences, enhanced operational efficiency, and predictive analytics capabilities, which allow real-time adaptation of marketing strategies to user preferences. These AI applications not only increase customer satisfaction and loyalty but also provide a significant competitive advantage in an increasingly demanding digital environment.

However, the research also reveals important ethical and technical challenges in AI implementation, such as algorithmic bias, model opacity, and privacy risks associated with managing personal data. These aspects can affect user trust and public perception of the company if not managed responsibly. In this regard, it is crucial for companies to adopt ethical and regulatory frameworks that promote transparency and accountability in AI use, protecting both brand integrity and consumer rights.

Additionally, this study contributes theoretically by consolidating existing literature and offering a comprehensive view of AI's impacts on corporate social networks. On a practical level, it serves as a guide for organizations and AI developers to understand both the opportunities and challenges inherent in implementing this technology on social media. Human oversight is particularly recommended at critical points to mitigate potential biases and ensure ethical decision-making in digital interactions. Establishing AI policies that promote fairness, transparency, and respect for

user privacy is essential to building a solid foundation of trust and sustainability in today's digital economy.

Finally, this study underscores the need for further research to deepen the understanding of AI applications across different sectors and cultural contexts, as well as the development of practical tools to audit and assess the ethical impact of these technologies in social networks. In this way, it is hoped that this research will serve as a foundation for a more ethical, transparent, and effective use of AI in brand communication within corporate social networks.

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Declarations

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